Ayush Kumar

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Professional Summary

Final-year B.Tech CSE (Data Science) student focused on building production-grade Generative AI systems and LLM-based applications.

Experienced in designing RAG pipelines, semantic retrieval systems, and deploying scalable AI services using FastAPI, Docker, and REST APIs. Strong foundation in embeddings, prompt engineering, and system-level optimization.

Education

B.Tech – Computer Science and Engineering (Data Science) Oct 2022 – Jun 2026
Haldia Institute of Technology (MAKAUT)
CGPA: 7.44 / 10

Technical Skills

Languages: Python (Flask, FastAPI, async basics), SQL

Generative AI: RAG Pipelines, Embeddings, Semantic Search, Prompt Engineering, LLM Evaluation

Agentic Frameworks: LangChain (basic), ReAct concepts, OpenAI APIs

Vector Databases: FAISS, PGVector (conceptual)

Backend & APIs: REST APIs, Microservices Basics

Cloud: AWS (S3, basic ECS/API Gateway exposure)

Data Engineering: ETL Pipelines, Schema Design, Data Validation, Query Optimization

DevOps: Docker, CI/CD (GitHub Actions), Git

Databases: PostgreSQL, MySQL, MongoDB

Concepts: System Design, Async Workflows

Experience

Finaure Ventures (Rupeia) — Data & AI Engineering Intern Jan – May 2025

- Built Python ETL pipelines processing **50K+ records daily**.
- Reduced manual processing time by **40%** via automated validation workflows.
- Designed schema validation and integrity checks improving reliability by **25%**.
- Optimized SQL queries across **100K+ rows**, reducing latency by **30%**.
- Implemented logging and monitoring for data drift and failure detection.
- Developed modular data processing components for reusable pipeline execution.
- Improved data quality by implementing anomaly detection and validation rules.
- Collaborated with cross-functional teams to support analytics and reporting systems.

Metamarkups — LLM Evaluation & AI Quality Intern Jul – Nov 2024

- Evaluated **3000+ LLM outputs** for hallucination, bias, and consistency.
- Designed regression-based evaluation pipelines across multiple tasks.
- Reduced critical errors by **15%** through iterative validation.
- Analyzed latency vs accuracy trade-offs for production decisions.
- Built structured evaluation frameworks for benchmarking model performance.
- Conducted prompt-level analysis to improve response quality and consistency.
- Supported testing of multiple model variants for real-world deployment scenarios.

Feynn Labs — Machine Learning Intern Jun 2023 – Aug 2023

- Built ML pipelines including preprocessing, feature engineering, and validation.
- Evaluated models using precision, recall, and F1-score metrics.
- Documented reproducible workflows and experiment tracking.
- Performed data cleaning and feature selection on structured datasets.

- Assisted in hyperparameter tuning and model performance optimization.
- Conducted exploratory data analysis (EDA) to extract insights from datasets.

Project

Financial News Analytics Platform (RAG + FastAPI + Docker)

- Developed a Retrieval-Augmented Generation (RAG) system to process and analyze 10K+ financial news articles.
- Implemented semantic search using sentence-transformer embeddings and FAISS for vector similarity.
- Designed a context-aware retrieval pipeline where user queries are converted into embeddings and matched with relevant document chunks.
- Integrated LLM-based response generation to provide accurate, context-grounded outputs, reducing hallucination.
- Built scalable REST APIs using FastAPI and deployed using Docker.
- Added logging, caching, and validation layers to improve system reliability and response accuracy.

AI Resume Screening System

- Built an AI-powered resume screening system using embedding-based similarity and rule-based scoring.
- Implemented hybrid ranking combining semantic similarity with weighted evaluation criteria.
- Designed explainable outputs to provide transparency in candidate ranking decisions.
- Developed modular backend APIs using FastAPI with database integration.
- Implemented evaluation pipelines to test ranking accuracy and consistency.

Snake Species Classification (Xception + CBAM)

- Developed a deep learning model using Xception architecture integrated with CBAM (attention mechanism).
- Trained on a dataset of 3000+ images across multiple snake species classes.
- Applied transfer learning and selective layer unfreezing to improve model performance.
- Achieved 86% validation accuracy through hyperparameter tuning and data preprocessing.
- Performed structured error analysis and benchmarking against baseline CNN models.

Achievements

- Secured **AIR 534 in GATE (Statistics)**, demonstrating strong foundation in probability, statistical inference, and mathematical rigor.
- Selected in the **Top 105 teams** (out of 3,700+ submissions and 64,000+ registrations) in the **Solution Challenge 2025** for building an AI-powered personalized learning platform (EduAI).
- Published research on **attention-enhanced deep learning (Xception + CBAM)** for multi-class image classification at MLSC-ES'24 seminar.